Statistical Learning and Data Analysis
Data Visualization

Emiliano del Gobbo
Statistics, tèchnè-loghia, analysis of data
DIETI, University of Napoli Federico II

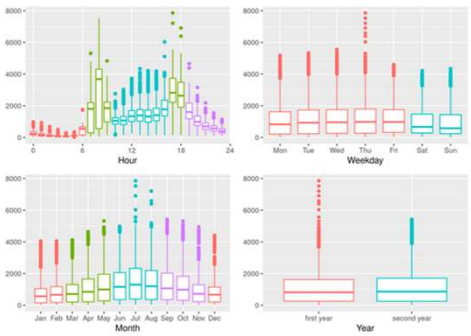
1

A thought

Start with the data, finish with a story!

In the middle there are a lot of plots and graphical analysis.

A graph is worth more than many words.



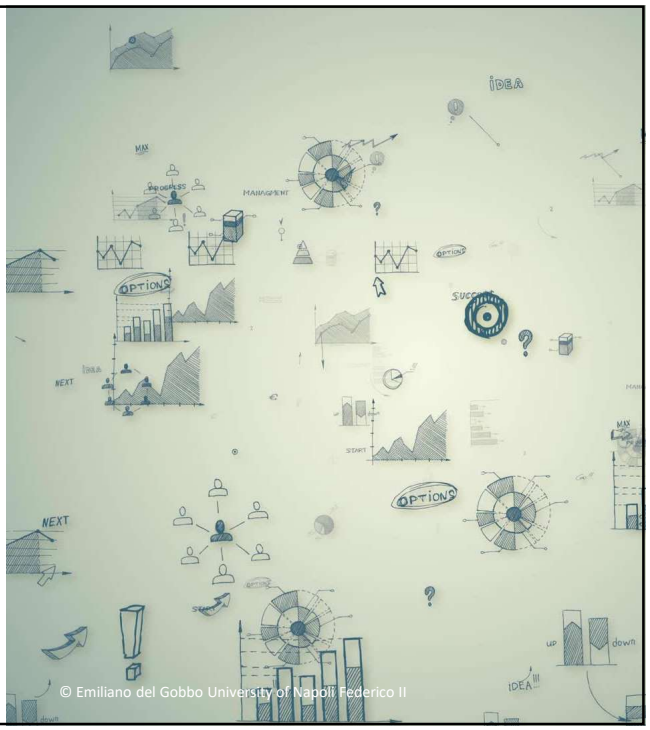
London bike sharing

© Emiliano del Gobbo University of Napoli Federico II

2

Data Visualization

- **Data and Information Visualization** refers to a broad set of methods and techniques to provide graphical representation of data and information in effective and comprehensible way.
- By using visual elements like charts, plots, graphs, maps, data visualization tools provide an accessible way to see and understand *trends, outliers, patterns, clusters in data.*
- *"There is no single statistical tool that is as powerful as a well-chosen graph"* in **"Graphical Methods for Data Analysis"** (Chambers, J.M., Cleveland, W.S., Kleiner, B., Tuckey, P.A., 1983), Wadsworth, Belmont.
- *Advanced data visualization tools are essential to analyze massive amounts of data and synthetize information to make data-driven decisions. These latter are presented in Data Mining and Machine Learning Module A Course.*



© Emiliano del Gobbo University of Napoli Federico II

Data Journalism

- Data visualization represents an excellent way for quantitative story-telling direct to technical and non-technical audience.
- **Data Journalism** refers to the increase role of data and summary quantification in the production and distribution of information in the digit era.
- The process is based on analyzing and filtering large data sets for the purpose to create a news story.
- ***Infographics (Information Graphics)*** *are visual representation of data, information and knowledge enhancing the human cognitive ability to see patterns and trends.*
- The steps **to raise value to public** are:

Data --> Filter --> Visualize --> Story

© Emiliano del Gobbo University of Napoli Federico II

Four Pillars of Data Journalism

Information

Story

Goal

Visual Design

© Emiliano del Gobbo University of Napoli Federico II

Titanic Case Study

- The sinking of the Titanic is one of the most infamous shipwrecks in history. On April 15, 1912, during her maiden voyage, the widely considered "unsinkable" RMS Titanic sank after colliding with an iceberg. Unfortunately, there weren't enough lifeboats for everyone on board, resulting in the death of 1502 out of 2224 passengers and crew. While there was some element of luck involved in surviving, it seems some groups of people were more likely to survive than others.
- The Kaggle challenge asks to build a predictive model that answers the question: *"what sorts of people were more likely to survive?"* using passenger data (i.e., name, age, gender, socio-economic class, etc).
- Exploratory Data Analysis (EDA) on the Titanic dataset helps to gain insights into the characteristics of passengers and their survival outcomes. Various types of data visualizations can be used to explore the data.
- https://www.kaggle.com/datasets/yasserh/titanic-dataset/data

© Emiliano del Gobbo University of Napoli Federico II

Understanding Titanic Data

https://www.kaggle.com/datasets/yasserh/titanic-dataset/data

- **Passenger ID** - To identify unique passengers
- **Survived** - If they survived or not
- **P Class** - The class passengers travelled in
- **Name** - Passenger Name
- **Sex** - Gender of Passenger
- **Age** - Age of passenger
- **SibSp** - Number of siblings or spouse
- **Parch** - Parent or child
- **Ticket** - Ticket number
- **Fare** - Amount paid for the ticket
- **Cabin** - Cabin of residence
- **Embarked** - Point of embarkment

© Emiliano del Gobbo University of Napoli Federico II

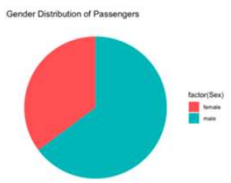
Pie Charts: How to interpret

A **pie chart** is a circular chart divided into sectors, illustrating relative magnitudes in frequencies or percent.

In a pie chart, the area is proportional to the quantity it represents.

- Slices (Segments): Each "slice" in a pie chart represents a category or subgroup of the whole dataset. The size of each slice is proportional to the percentage or count of data points within that category relative to the entire dataset.
- Angle of Slices: The angle of each slice at the center of the pie is determined by the proportion it represents. Larger angles correspond to larger proportions. The sum of all the angles in the pie chart is always 360 degrees, representing the entire dataset.
- Labels and Legend: Categories are typically labeled either within the slices or in a legend adjacent to the pie chart. Labels show the category names and may include the percentage or count represented by each slice.

Because the human eye is good at judging linear measures and poor at judging relative areas, a **bar plot** is often preferable to pie charts as a way to display categorical data.



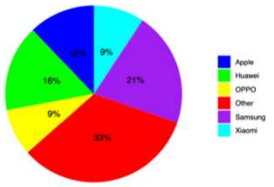
Titanic data set

© Emiliano del Gobbo University of Napoli Federico II

Pie Charts: where to use

- ✓ **Composition:** Pie charts are useful to visualize the relative proportions of different categories in a dataset, making it easy to understand the composition of a whole in a visually intuitive manner.
- ✓ **Comparison:** Pie charts are excellent for comparing the relative sizes of categories within the dataset, identifying which categories dominate the distribution and which are less prominent.
- ✓ **Highlighting:** One can emphasize specific slices by "exploding" or pulling them out from the center of the pie chart. This is done to draw attention to a particular category.
- ○ Pie charts should be ***avoided when there are too many categories***, making the chart hard to read. Additionally, pie charts may not be the best choice when comparing categories ***with small differences in proportions***.

Data Source: Ensure that you are aware of the source of the data and any relevant units or scales used in the chart.



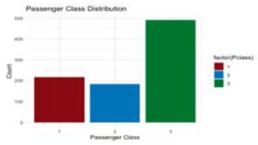
Example: Fake data showing global market share for mobile phone manufacturers.

© Emiliano del Gobbo University of Napoli Federico II

Bar Plot: How to interpret

A **bar chart** associates each distinct category of the variable with a rectangular bar whose height is proportional to the absolute or relative frequency.

- **Understanding the Axes:** Vertical Axis typically represents the frequency or count of each category. It shows how many data points fall into each category. Horizontal Axis displays the categorical variables or labels. Each bar corresponds to a specific category on this axis.
- **Bar Length:** The length of each bar on the chart represents the quantity or count associated with the corresponding category on the X-axis. Taller bars indicate a higher count or frequency.
- **The choice to use absolute or relative frequencies** does not change the shape of the graph. The use of relative frequencies is a must when you want to make comparisons of the same character observed on collectives of different numbers (for example, how the opinions of customer satisfaction are distributed in two accommodation facilities in comparison).
- **Labels and Titles:** Pay attention to the chart's title and axis labels. They provide context and help you understand what the chart is representing.
- **Bar Colors (if applicable):** Sometimes, bar charts use colors to distinguish between categories or highlight specific points of interest. The legend, if present, explains the meaning of the colors.



Titanic data set

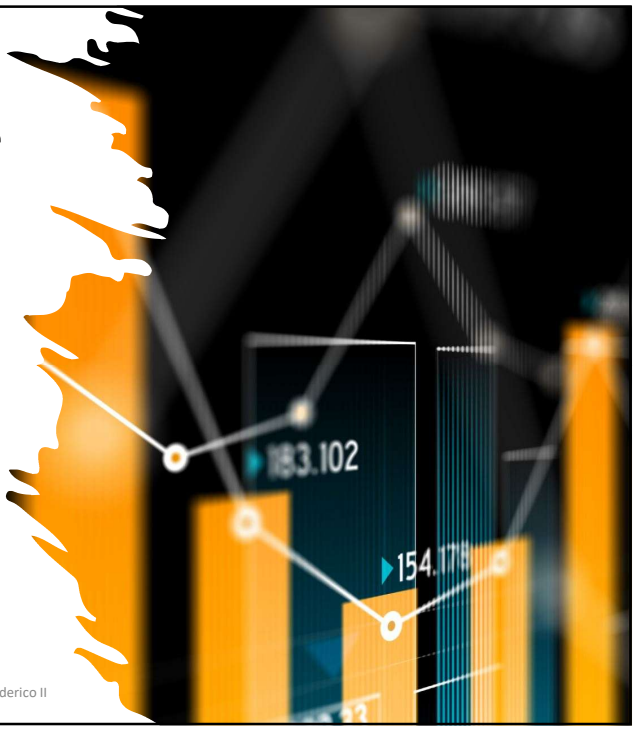
The first bar plot appeared in a 1786 book The Commercial and Political Atlas, by William Playfair (1759-1823), a pioneer in the use of graphical data visualization techniques.

© Emiliano del Gobbo University of Napoli Federico II

Bar Plots: Where to use

- ✓ **Comparison:** Bar charts are useful for visualizing the distribution of categorical data, comparing the sizes of different categories, and identifying which categories have the highest or lowest counts by comparing the lengths of the bars.
- ✓ **Patterns in the distribution:** Observe the distribution of data across categories. Are there categories that dominate the chart, or is the distribution relatively even?
- ✓ **Outlier detection:** Look for any categories that stand out as outliers with significantly higher or lower counts than the others.
- ○ *Bar charts are less effective when dealing with continuous data or when the number of categories is extensive, as too many bars can clutter the plot.*

Data Source: Ensure that you are aware of the source of the data and any relevant units or scales used in the chart.



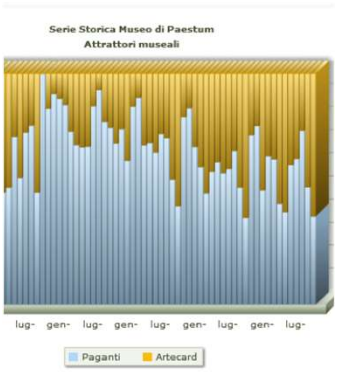
© Emiliano del Gobbo University of Napoli Federico II

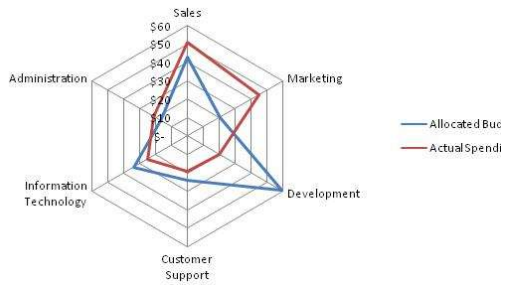
Compositional Data Visualization

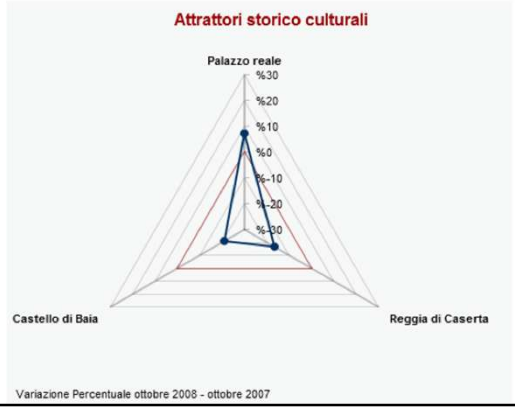
Compositional Data are obtained considering the percentage distribution of a total frequency or intensity for the distinct modalities of a variable.

The idea is to use a bar in which the different dimensions of the variable can be distinguished with different colors.

Here is an example in which the monthly variation of the division between Paying and Non-Paying is shown in a museum attractor such as the archeological site of Paestum.







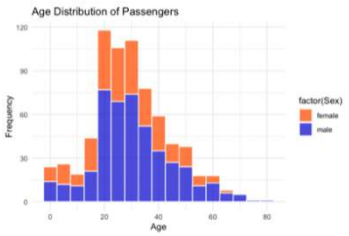
Radar or Spider Chart

The **Radar Chart** for a categorical variable with three distinct modalities.

In the bottom example, the chart represents the different levels in which the compositional data is divided between the three distinct modalities.

A comparison is made in terms of percentage variation between one year and another to highlight the dynamics of the phenomenon over time.

© Emiliano del Gobbo University of Napoli Federico II



Titanic data set

Histogram: How to interpret

- Similar to a bar chart, **histogram** is for continuous variables where area of any bar is proportional to the frequency of units in the bar.
- **The range is partitioned into intervals.**
- The **y-axis** can be
  - Frequency (heights add up to n) for *equi-width intervals*
  - Percentage (heights add up to 100%) for *equi-width intervals*
  - **Relative Density** (Areas add up to 1) for *different-width intervals*
- Relative Density= $\frac{\text{Relative Frequency}}{\text{Interval Range}}$

© Emiliano del Gobbo University of Napoli Federico II

Histograms: Where to use

Histograms are effective for visualizing the distribution of **continuous variables**, identifying data patterns, and understanding data skewness and central tendency.

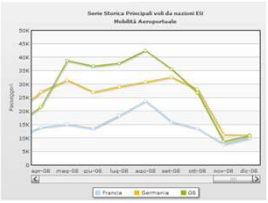
- ✓ **Central Tendency:** the central peak (if present) in a histogram indicates the mode or the most frequently occurring value within the data. The center of mass of the histogram, which may or may not coincide with the mode, provides information about the data's central tendency.
- ✓ **Spread and Dispersion:** The spread of the histogram, determined by the width and shape of the distribution, reflects the variability or dispersion of the data. Wider distributions suggest higher variability, while narrower distributions suggest lower variability.
- ✓ **Outliers:** Histograms can help identify outliers, say values that fall far from the central mass of the data. Outliers are often displayed as bars separate from the main distribution.
- ✓ **Skewness:** The direction in which a histogram tilts, either to the left (negative skew) or right (positive skew), indicates the skewness of the data distribution.
- ✓ **Data Range:** The range of values on the X-axis represents the minimum and maximum values within the dataset.
- ○ *Histograms are not suitable when dealing with categorical data or when the data distribution is heavily skewed, as they may not accurately represent the underlying patterns.*

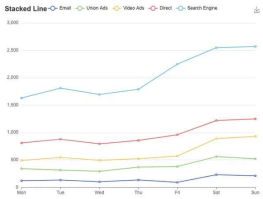
Data Source: Ensure that you are aware of the source of the data and any relevant units or scales used in the histogram.

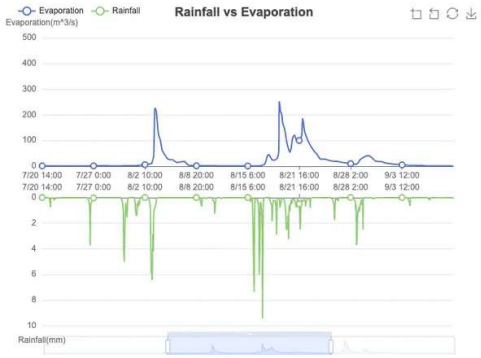
© Emiliano del Gobbo University of Napoli Federico II

Time Series Visualization

The temporal data are the values of a numeric character observed at different instants of time (i.e., day, month, quarter, semester, year).





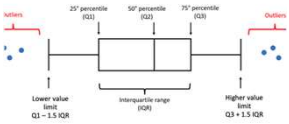


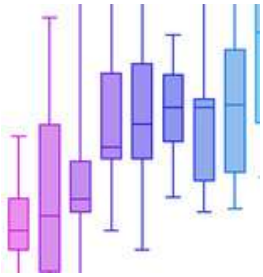
© Emiliano del Gobbo University of Napoli Federico II

Box-Plot (John Wilder Tukey 1950)

Box and Whisker Plot

- **Box-Plot** is a very efficient summary of numerical variables visualizing locality, spread and skewness:
  - The **box width** is equal to the **IQR** obtained by the difference between the Third Quartile and the First Quartile,
  - The **line in the middle** is the **Median**,
  - The **Whiskers** extending at **1.5 IQR** from the box indicating variability outside the first and third quartiles.
  - **Outside the whiskers range** are denoted **outlier points** with unusual values.
- Box-Plot demonstrates skewness, not bimodality.
- It is very useful to compare the distribution of a numerical variable in different groups of units (i.e., time, space, levels of a stratifying factor).





© Emiliano del Gobbo University of Napoli Federico II

Bike Sharing Case Study

- Bike sharing is a service that allows people to temporarily and flexibly rent bikes. Users can go to docking stations, take a bike, ride it and then return it to a different docking station. Bike sharing is nowadays very practical since the entire process of customer membership, rental and return is automatically managed by computers, that can incidentally collect data about every aspect of the rides, i.e., time, duration and so on. These systems are growing because they address a variety of problems such as city traffic, pollution and personal health.
- The Challenge focus on the number of bike rentals (bike count) as target variable, and *its relation to date-time and weather data, that will be treated as explanatory or predictor variables.*
- Data downloaded from Kaggle (London bike sharing dataset) and it is an aggregate of bike sharing data from the "TfL Open Data", the London public transport service and weather data from freemeteo.
- It is a time-series of 17414 hourly measurements of the number of bike rentals, weather data, and type of day, over the span of two years, from the 4th of January 2015 to the 3rd of January 2017, for a total of 11 variables.
- More specifically weather measurements are: temperature, humidity, wind speed, and the weather code which can be one of the following: mostly clear, scattered clouds, broken clouds, cloudy, rain, rain with thunderstorm, snowfall and freezing fog.
- About the day it is specified whether it is a holiday, a weekend or not, and the corresponding season.
- The data set has no missing values. The names of the variables and categories were modified to be clearer, and the time-stamp was separated in date and time, see table for a summary.

Table 1: Variables in the data set

Name	Type	Unit
`date`	Discrete	ymd
`time`	Discrete	hms
`bike_count`	Discrete	
`temperature`	Continuous	°C
`humidity`	Continuous	%
`wind_speed`	Continuous	km/h
`weather_code`	Categorical	
`season`	Categorical	
`is_holiday`	Boolean	
`is_weekend`	Boolean	

© Emiliano del Gobbo University of Napoli Federico II

Hourly	bike_count	humidity	temperature	wind_speed
Skewness	1.33	−0.57	0.20	0.67
Kurtosis	1.54	−0.26	−0.26	0.45
Pearson χ^2	20431.85	20684.84	35406.6	36526.09
p-value	0	0	0	0

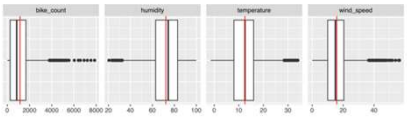


Figure 1: Boxplots of numerical variables, red lines show the average values.

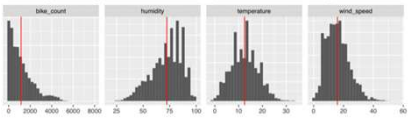


Figure 2: Histograms of numerical variables, red lines show the average values.

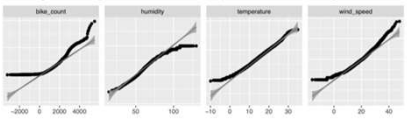


Figure 3: QQ plots of the numerical variables.

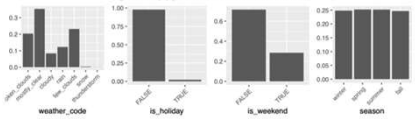


Figure 4: Barplots of categorical and Boolean variables.

© Emiliano del Gobbo University of Napoli Federico II

Time Box-Plots

- The effect of strikes in London bike sharing.
- Data downloaded from Kaggle (London bike sharing dataset) and it is an aggregate of bike sharing data from the "TfL Open Data", the London public transport service and weather data from freemeteo.

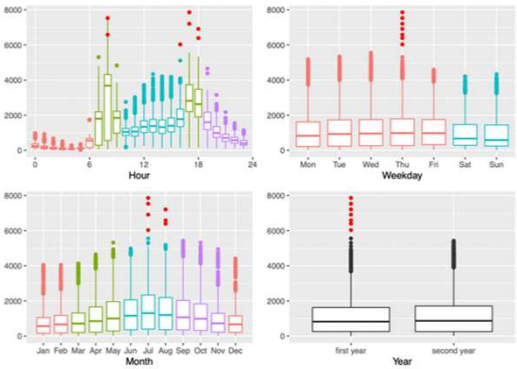
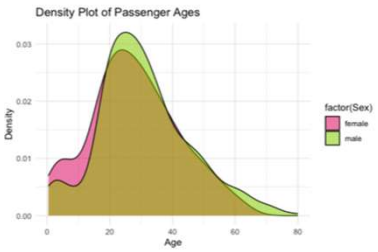


Figure 5: Boxplots of `bike_count` across hour, weekday, month and year. In the hourly plot, colours show the following groups: night (0 to 6), rush hour (7 to 9 and 17 to 18), afternoon (10 to 16) and evening (19 to 24). In the weekday plot they show working days and weekend. In the monthly plot they show the season: winter, spring, summer and fall. Red dots show the number of bikes per hour over 6000, that only happened during subway strikes.

© Emiliano del Gobbo University of Napoli Federico II

Density Plot: How to interpret



A **density plot** is a smoothed representation of the distribution of continuous data.

It provides a continuous curve that approximates the data distribution.

The idea is to use sample data to infer the **Probability Density Function (PDF)** which is *the model generator of our data. We move in the Induction/Inference Approach!*

Density plots are useful for visualizing the data's distribution, including patterns, peaks and modes, spread.

The density estimation process smooths out the noise in the data, revealing underlying patterns.

The smoothed curve is also known as Kernel Density Estimate.

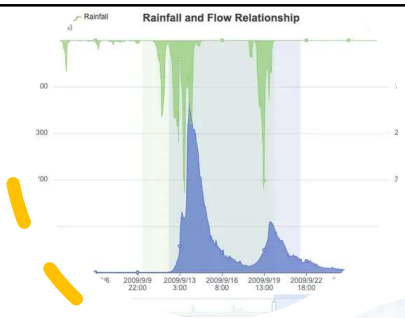
The **height of the curve** at any point indicates the estimated density of data points near that value.

The **area under the curve** between two points represents the probability that a randomly selected data points falls within that range.

Integrating the curve across the entire range equals 1, as it represents the entire dataset.

© Emiliano del Gobbo University of Napoli Federico II

Density plot: more about interpretation



Peaks or modes in the density plot represent regions of higher data density. *Each peak* corresponds to a mode in the data distribution.

- **Multimodal distributions** have *multiple peaks*, indicating different groups or clusters within the data.

Spread and Variability: The width of the density plot curve at any point reflects the spread or variability of data points around that value.

- Wider sections indicate higher data variability, while narrower sections suggest lower variability.

Skewness and Symmetry: Density plots can help identify skewness in the data distribution.

- *Positive skew* results in a long tail to the right, while *negative skew* has a long tail to the left.
- Symmetric distributions have a balanced, *bell-shaped curve*.

Comparison: Density plots can be used to compare the distributions of different datasets or subgroups. Overlapping or distinct curves reveal differences in distribution characteristics.

Data Source: Ensure that you are aware of the source of the data and any relevant units or scales used in the density plot.

© Emiliano del Gobbo University of Napoli Federico II

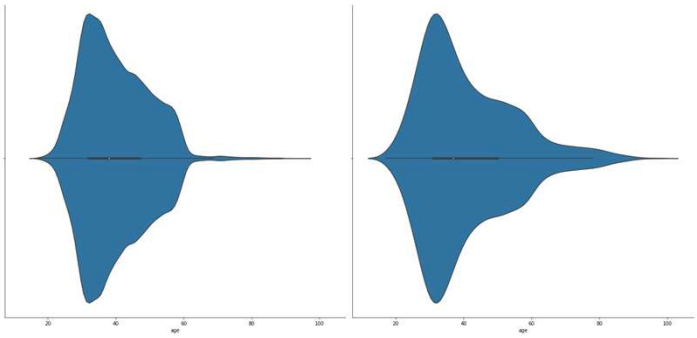
Violin Plot

- ✓ The **density of the plot** reflects the concentration of data points at that value. *Higher density suggests a greater number of data points in that region.*
- ✓ **Thick line or dot inside** the violin represents the median or mean of the data for each category.
- ✓ *Wider sections indicate higher variability, while narrower sections suggest lower variability*

© Emiliano del Gobbo University of Napoli Federico II

It is similar to box-plot, it adds a rotated *kernel density plot.*

$Y = 0\ (NO)$ $\qquad$ $Y = 1\ (YES)$



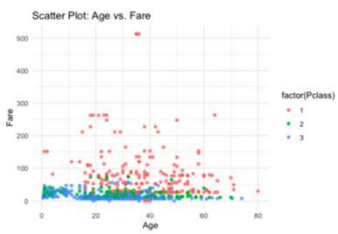
Real world case study: *Is it possible for a Bank to know if a client will subscribe a term deposit during a marketing campaign?*
Target variable $Y = 0,1$*: NO/YES*

Scatter Plot: Where to use

- A **scatter plot** is a two-dimensional visualization that displays individual data points as dots on a graph, with one variable on the X-axis and another on the Y- axis. It is used to visualize the relationship or correlation between two continuous variables, two detect patterns and identify outliers.
- **Variables and Axes:** A scatter plot displays data points as individual dots on a two-dimensional factorial representation. One variable is typically plotted on the X- axis, and the other on the Y- axis.
- **Each dot** represents a single data point, with its position determined by the values of the two variables.
- **Axes Labels and Units:** Pay attention to the labels on the X- axis and Y- axis as well as the units of measurement.
- **Scales:** Understanding the scales is crucial for interpreting the relationship correctly.

✓ Appropriate scenarios: Scatter plots are valuable when exploring the relationship between two continuous variables, identifying trends, patterns, or outliers, and assessing the strength and direction of the correlation.

○ When not to use scatter plots: Avoid using scatter plots when dealing with categorical variables or when exploring more than two variables simultaneously, as scatter plots are limited to two dimensions.



Titanic data set

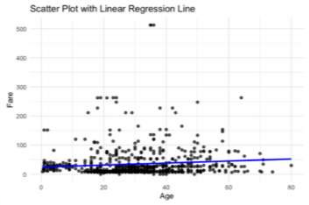
© Emiliano del Gobbo University of Napoli Federico II

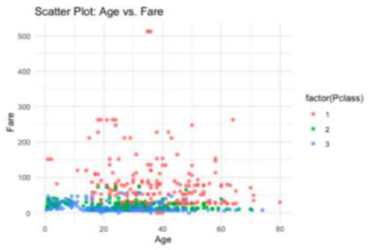
Scatter Plot: How to interpret

Key aspects to understand the nature of the relationship:

- ✓ **Direction:** Determine whether there is a positive or negative relationship. Positive relationships show an upward trend, while negative relationships show a downward trend.
- ✓ **Strength:** Assess the strength of the relationship. A strong relationship is characterized by closely clustered points, while a weak relationship results in more scattered points.
- ✓ **Linearity:** Check for linearity, as linear relationships are the simplest to analyze. Nonlinear relationships may exhibit curves or clusters.
- ✓ **No Relationship:** In some cases, a scatter plot may show no discernible relationship, with points scattered randomly.
- ✓ **Outliers:** Identify any data points that fall far from the main cluster of dots. These outliers may indicate extreme or unusual values.
- ✓ **Patterns and Clusters:** Look for patterns, clusters, or groupings of data points that suggest subpopulations or distinct trends within the data. Clusters may indicate different categories or behaviors within the dataset.
- ✓ **Data Density:** Assess the density of data points within specific regions of the scatter plot. Higher data density indicates a concentration of values in that area.

Data Source: Ensure that you are aware of the source of the data and any relevant context, such as the purpose of the analysis or any known patterns.



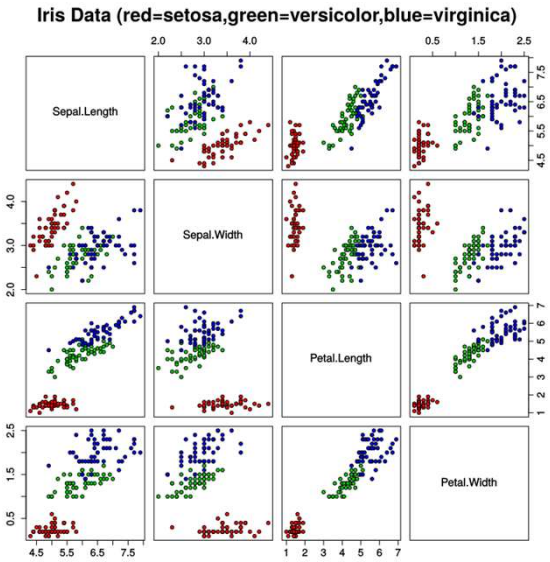


Titanic data set

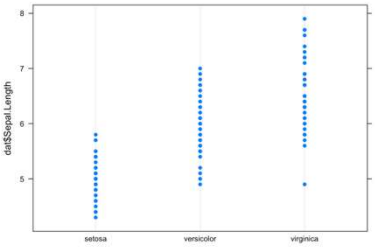
© Emiliano del Gobbo University of Napoli Federico II

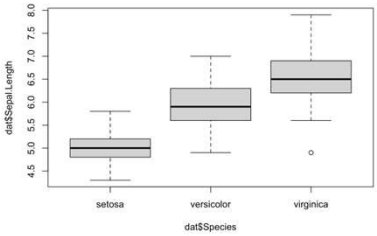
Scatterplot Matrix of Fisher's Iris data set (1936)

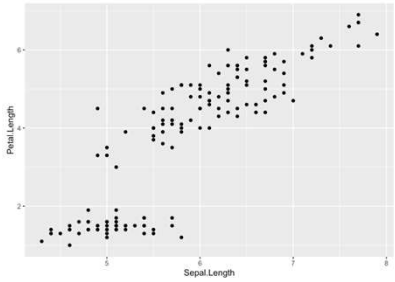
- Iris flower data set or Fisher's Iris data set (1936):
  - 50 samples from each of three species of Iris (Iris setosa, Iris virginica, Iris versicolor)
  - 4 features, i.e., the length and the width of the sepals and petals, in cm.
- Fisher developed a linear discriminant model to distinguish the species from each other on the basis of the features' measurements.



© Emiliano del Gobbo University of Napoli Federico II



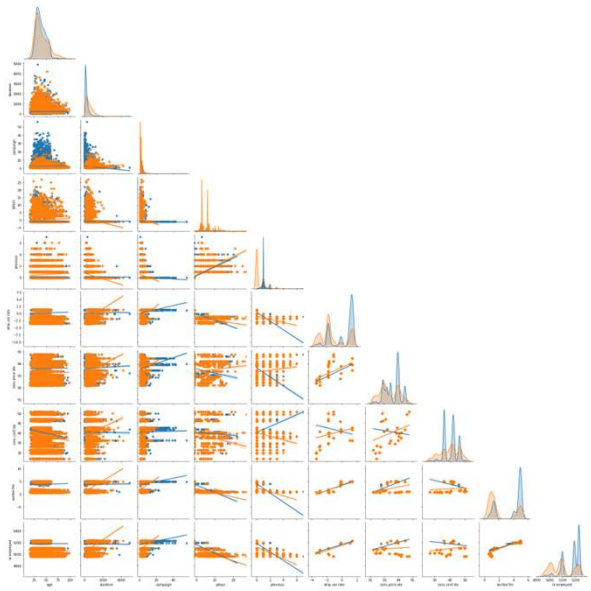




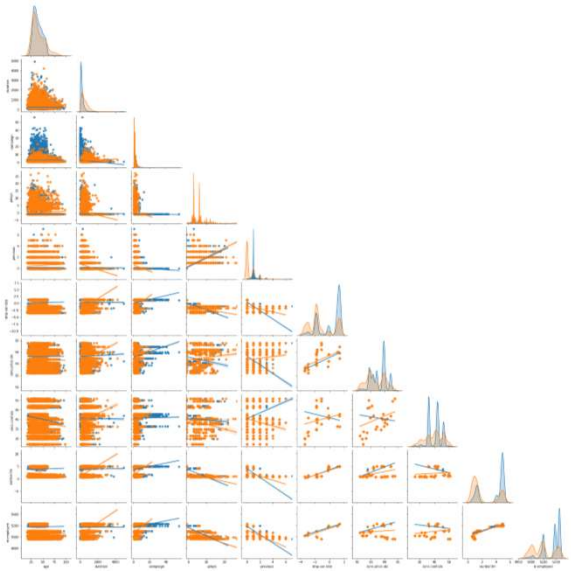
Dot Plot, Box-Plot, Scatter Plot (Iris Data Set)

© Emiliano del Gobbo University of Napoli Federico II

Pair Plot



© Emiliano del Gobbo University of Napoli Federico II



Scatter Plot Matrix or Pair Plot: Where to use

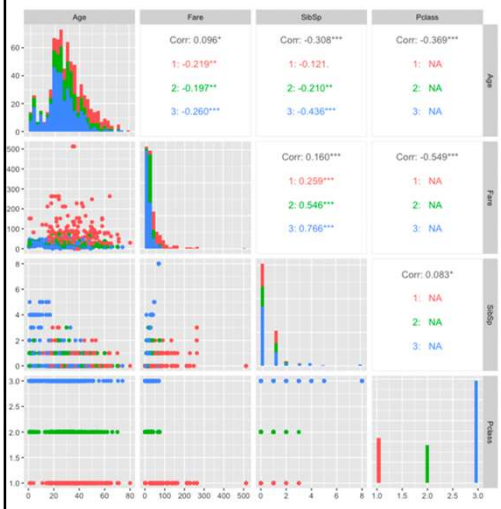
A **Scatter Plot Matrix** or **Pair Plot** is a comprehensive visualization that displays scatter plots or pairwise relationships between multiple variables in a dataset.

- **Diagonal Plots (Histograms/Density):** The diagonal cells of the Pair Plot display density plots (or histograms) for each individual variable, providing insights into their univariate distributions.
- **Scatterplots (Off-Diagonal):** In the off-diagonal cells of the matrix, Pair Plots are used to visualize the relationship between pairs of variables.
- *Each point in a pair plot represents a data point, with one variable on the X-axis and another on the Y-axis.*

✓ **Suitable scenarios**: Pair plots are highly effective when exploring the relationships between multiple numeric variables simultaneously. They help identify patterns, trends, and correlations within the data.

○ *When to consider other visualization methods: Pair plots can become cluttered and less informative when dealing with a large number of variables or when working with categorical variables.*

© Emiliano del Gobbo University of Napoli Federico II



Titanic data set stratified by the factor PClass - "Passenger Class"

Scatterplot Matrix or Pair Plot: How to interpret

- A **Scatter Plot Matrix**, also known as a **Pair Plot**, is a comprehensive visualization that displays scatter plots for each pair of variables in a matrix format, allowing us to assess correlations and patterns.
- **Correlation Coefficients (Optional):** Some pair plots include correlation coefficients, typically Pearson's linear correlation coefficient to quantify the strength and direction of linear relationships. A correlation coefficient close to 1 or -1 indicates a strong linear relationship, while values close to 0 suggest weak or no linear relationship.

Key aspects to understand the nature of the relationship:

✓ **Linear Relationships:** Look for patterns that suggest linear correlations. Positive correlations show an upward trend, while negative correlations show a downward trend.

✓ **Non-linear Relationships:** Identify nonlinear patterns, such as curves or clusters, that may indicate more complex relationships.

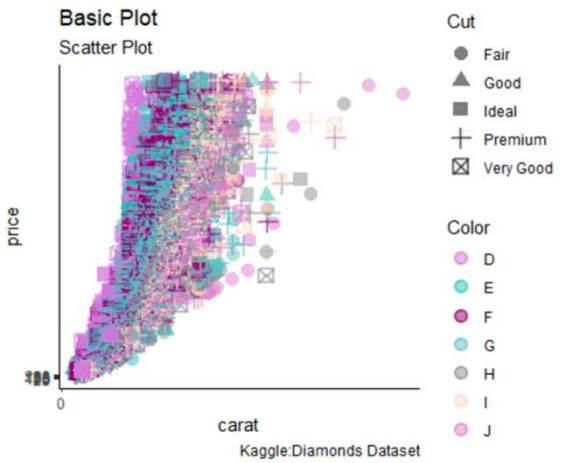
✓ **No Relationship:** In some cases, scatterplots may reveal no discernible relationship between variables, appearing as scattered points with no clear trend.

- **Patterns and Clusters:** Look for patterns, clusters, or outliers in the scatterplots that may suggest subgroups or specific trends within the data.

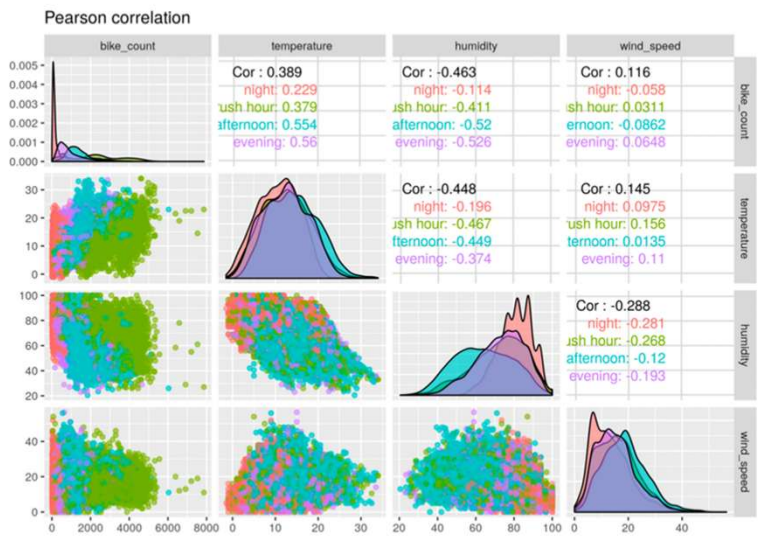
Data Source: Ensure that you are aware of the source of the data and any relevant units or scales used in the pair plot.

© Emiliano del Gobbo University of Napoli Federico II

Scatter Plot with stratifying factors

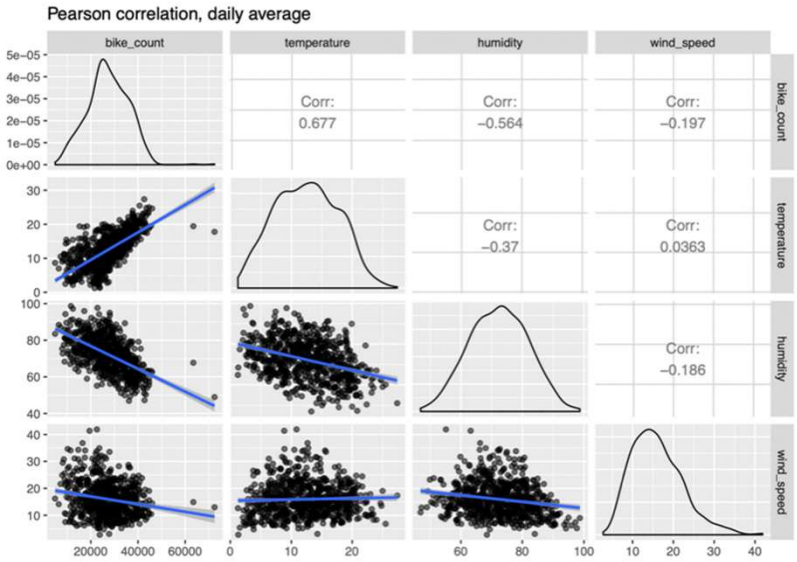


© Emiliano del Gobbo University of Napoli Federico II



London bike sharing Case Study: Pearson linear correlation across numerical variables, hourly measurements.

© Emiliano del Gobbo University of Napoli Federico II



London bike sharing

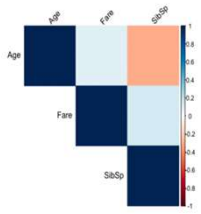
© Emiliano del Gobbo University of Napoli Federico II

Heat Map: Where to use

- A **Heatmap** is a graphical representation of data in a matrix format, where colors are used to depict the magnitude or relationships between variables.
- *Heatmaps are commonly used to visualize patterns or correlations in multivariate datasets.*

✓ **Suitable situations:** Heatmaps are effective for visualizing relationships, patterns, and correlations among multiple variables in tabular or matrix-like data.

○ **Limitations and constraints:** Heatmaps may become less informative when dealing with very large datasets or when the number of variables is extensive.



Titanic data set

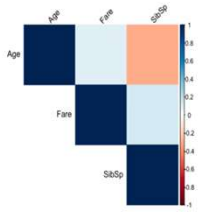
© Emiliano del Gobbo University of Napoli Federico II

Heat Map: How to interpret

Interpreting a correlation heatmap is essential for understanding the relationships and dependencies between multiple continuous variables in a dataset. Correlation heatmaps are particularly useful for identifying patterns, strengths, and directions of linear relationships. Here's how to interpret a correlation heatmap effectively:

- **Heatmap Color Scale:** In a correlation heatmap, colors are used to represent the strength and direction of correlations. *A color scale is typically provided alongside the heatmap*. The color scale often ranges from a cool color (e.g., blue) representing negative correlations to a warm color (e.g., red) representing positive correlations. Neutral or no correlation is often represented by a color close to white.
- **Correlation Values:** The values within each cell of the heatmap represent correlation coefficients. Commonly used correlation coefficients include Pearson's linear correlation coefficient for linear relationships. Correlation values range from -1 to 1, with -1 indicating a perfect negative correlation, 1 indicating a perfect positive correlation, and 0 indicating no linear correlation. The closer the value is to -1 or 1, the stronger the correlation. Values close to 0 suggest weak or no linear relationship.
- **Strenght of Correlations:** Identify cells with strong correlations (high absolute values). These represent variables that have a significant linear relationship. Weak or near-zero correlations suggest that the variables have little to no linear relationship.
- **Direction of Correlations**: Observe the sign (positive or negative) of the correlation coefficients. Positive values indicate a positive linear relationship, while negative values indicate a negative linear relationship. The direction of the relationship is important for understanding how variables change together.
- **Heatmap Structure:** Look for patterns and structures within the heatmap. Clusters of variables with similar correlations may indicate subgroups or related features. High positive or negative correlations between multiple variables may suggest redundancy or multicollinearity.
- **Thresholds and Significance:** Consider setting correlation thresholds to focus on relationships above a certain strength. For example, you may choose to highlight correlations with absolute values above 0.5 as significant. Assess the statistical significance of correlations, especially in smaller datasets, to ensure that observed relationships are not due to chance.

Data Source: Ensure that you are aware of the source of the data and any relevant context or domain knowledge when interpreting correlations.



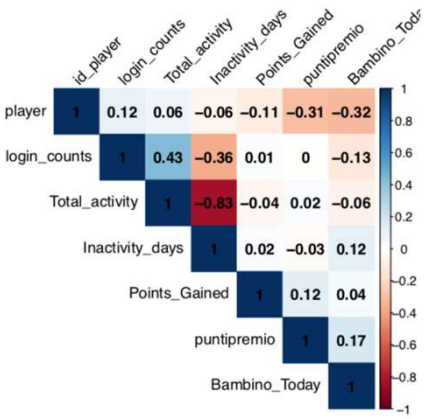
Titanic data set

© Emiliano del Gobbo University of Napoli Federico II

Heat Map: How to interpret

Interpreting a correlation heatmap is essential for understanding the relationships and dependencies between multiple continuous variables in a dataset. Correlation heatmaps are particularly useful for identifying patterns, strengths, and directions of linear relationships. Here's how to interpret a correlation heatmap effectively:

- **Heatmap Color Scale:** In a correlation heatmap, colors are used to represent the strength and direction of correlations. *A color scale is typically provided alongside the heatmap*. The color scale often ranges from a cool color (e.g., blue) representing negative correlations to a warm color (e.g., red) representing positive correlations. Neutral or no correlation is often represented by a color close to white.
- **Correlation Values:** The values within each cell of the heatmap represent correlation coefficients. Commonly used correlation coefficients include Pearson's linear correlation coefficient for linear relationships.



Fater Challenge Case Study

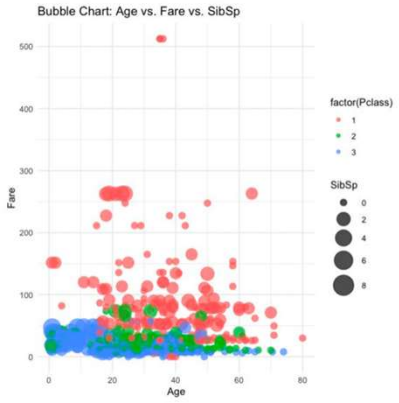
© Emiliano del Gobbo University of Napoli Federico II

Bubble Chart

- A **Bubble Chart** is a variation of a scatter plot where the size of the bubbles represents a third variable. It is used to visualize three-dimensional data by using position on the x and y axes for two variables and the size of the bubbles for the third variable.

✓ Effective usage of bubble charts: Bubble charts are valuable for visualizing three- dimensional data, allowing you to explore the relationships between two continuous variables while incorporating a third variable through bubble size.

○ When to consider other visualization methods: Consider alternative visualizations when dealing with more than three variables, as bubble charts may become too complex.

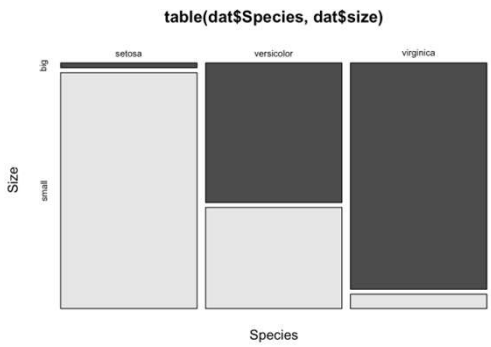


Titanic data set

© Emiliano del Gobbo University of Napoli Federico II

Mosaic Plot

- A **Mosaic Plot** allows to visualize a contingency table which cross-classify two categorical/categorized variables.
- Frequencies correspond to the areas:
  - the areas of the rectangular tiles corresponding to a combination of categories are proportional to the number of observations that have that combination of categories.
- Three-way tables are also possible to visualize, higher dimension becomes confusing.

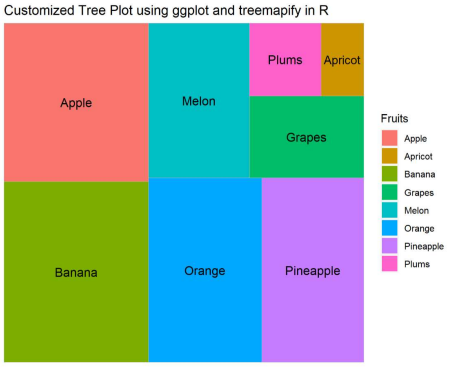


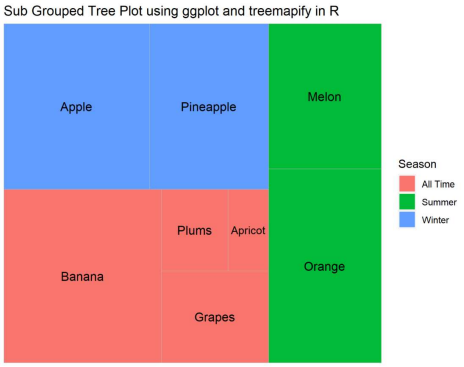
Iris Data Set

© Emiliano del Gobbo University of Napoli Federico II

TreeMap

- A **TreeMap** is a hierarchical visualization that represents data as nested rectangles. It is used to visualize hierarchical structures and part-to-whole relationships within data.



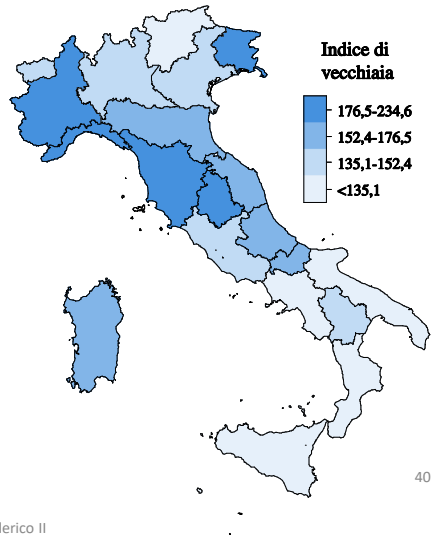


© Emiliano del Gobbo University of Napoli Federico II

Territorial series

- They are often represented by **cartograms**: territorial divisions are shown on the appropriate map, and the corresponding intensities (frequencies or quantities) are represented using different colors or shading patterns, whose meanings are specified in dedicated legends.
- A variant is the **cartodiagram**, in which each territorial division is associated with a chart representing the distribution or levels of one or more phenomena.

cartogram of the aging index by region



© Emiliano del Gobbo University of Napoli Federico II

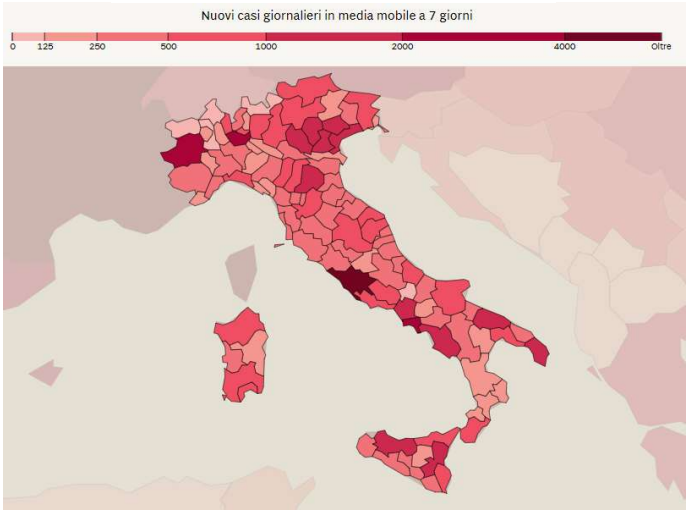
Territorial series

Cartodiagram

© Emiliano del Gobbo University of Napoli Federico II

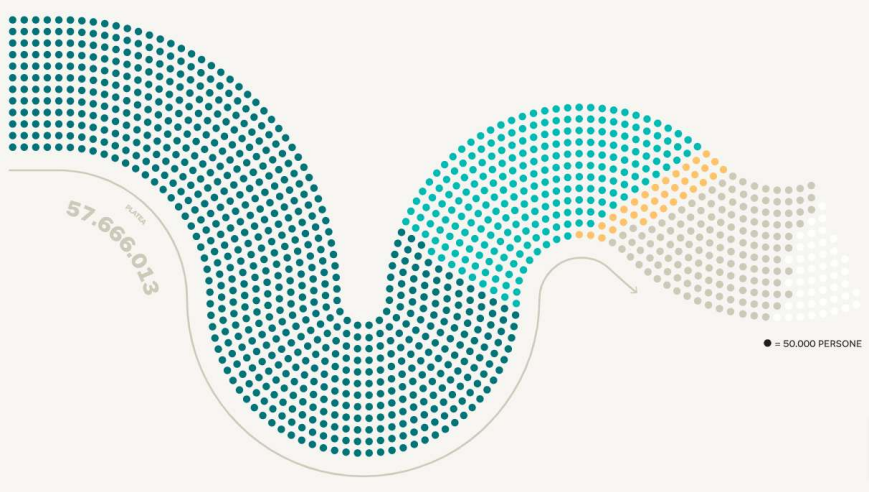
Territorial series

New case of Covid



© Emiliano del Gobbo University of Napoli Federico II

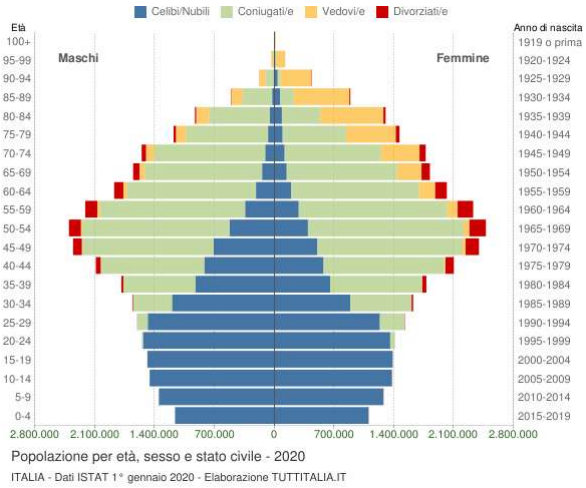
Alternative visualizations



Covid Vaxination in Italy

© Emiliano del Gobbo University of Napoli Federico II

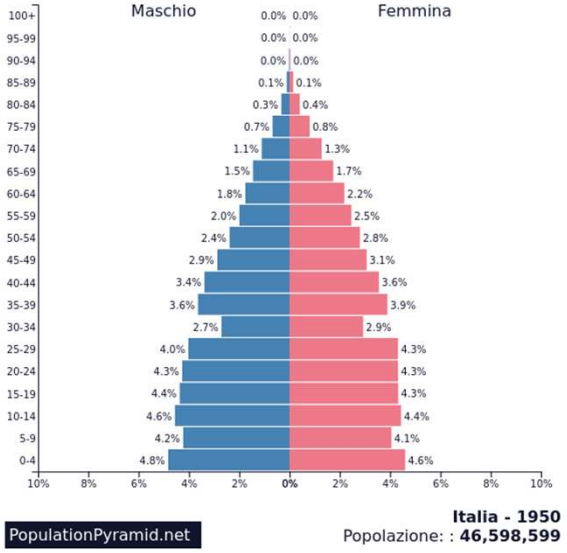
Population pyramid



© 2018 - Emiliano del Gobbo

© Emiliano del Gobbo University of Napoli Federico II

Population pyramid



© Emiliano del Gobbo University of Napoli Federico II

Scale problems

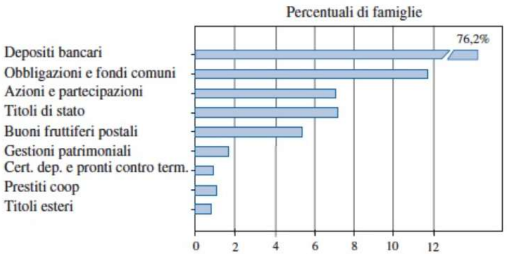
- When the quantities represented show a high degree of dispersion, with values that differ greatly, the issue of **scaling** arises.
  If the scale is adjusted to fit the smaller values, the larger ones become compressed.
  Conversely, if the scale is based on the larger values, the smaller ones become unreadable.

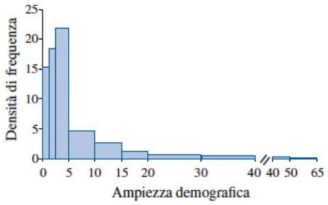
Possible solutions:

- Indicate to the reader that in some sections the scale is not consistent
- Apply a change of scale

© Emiliano del Gobbo University of Napoli Federico II

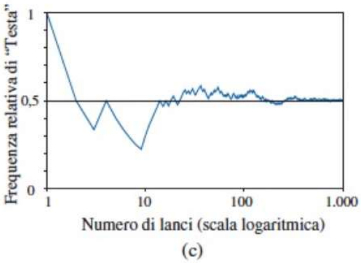
Ad hoc solutions

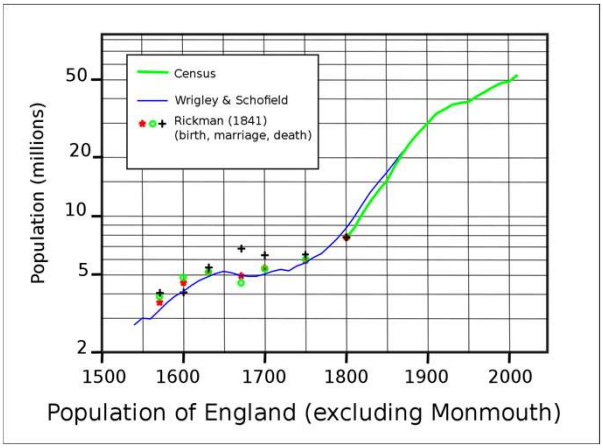




© Emiliano del Gobbo University of Napoli Federico II

Change of scale





© Emiliano del Gobbo University of Napoli Federico II